LinkSnip

Smart URL Shortening with Click Analytics

A full-stack web application · FastAPI + React + SQLite

Presented by

Shaswati Sahoo

OVERVIEW

Turning long links into measurable insight

THE CHALLENGE

Long, messy URLs are hard to share and impossible to measure. Once a link is out in the world, you have no idea who clicked it, from where, or on what device.

THE SOLUTION

LinkSnip shortens any URL in one click — with optional custom aliases and expiry rules — and turns every click into actionable analytics through a clean, secure dashboard.

Shorten & Brand
Custom short links & aliases

Track Every Click
Real-time click analytics

Own Your Data
Private, per-user dashboards

CAPABILITIES

Key Features

URL Shortening
Instantly shorten any link, with optional custom aliases.

Link Expiry
Auto-expire links by date or by a maximum click limit.

Click Analytics
Track clicks over time, top referrers, devices & browsers.

Geo Insights
See where clicks come from on an interactive world map.

QR Codes
Generate a scannable QR code for any short link.

Accounts & Auth
Secure JWT signup / login with a personal dashboard.

UNDER THE HOOD

Technology Stack

Frontend
React 19 · TypeScript · Vite · React Router

Backend
FastAPI · Uvicorn (async Python REST API)

Data Layer
SQLModel ORM · SQLite (zero-config database)

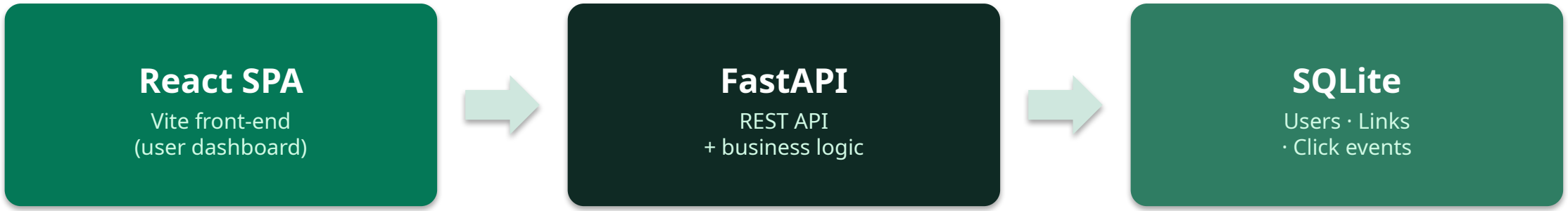
Auth & Security
JWT (PyJWT) · passlib password hashing

Analytics
geoip2fast geolocation · click logging

UX Extras
QR code generation · D3 world-map visualization

HOW IT WORKS

Architecture at a glance



THE REDIRECT FLOW

Short link click → 302 redirect to the destination → **click logged** (timestamp · referrer · device · browser · country)

A 302 (not 301) is used on purpose — so **every click reaches the server and is counted**, never cached away by the browser.

ENGINEERING

Technical Highlights

1. **Base62 short codes** — derived from row IDs — unique by design, zero collision checks.
2. **302 redirects, not 301** — so every click is counted and never cached away.
3. **Geo + device analytics** — captured on each redirect via GeoIP and user-agent parsing.
4. **JWT-secured, per-user data** — strict 401 / 404 ownership guards protect every link.
5. **Idempotent auto-migrations** — the schema evolves on startup without losing data.
6. **Clean API design** — stored models vs. exposed schemas kept cleanly separated.

Thank You

LinkSnip — Smart URL Shortening & Analytics

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